

12 FOOD

With more and more humans crowding for space on the planet, food production is going to be an important aspect that we need to get right, or else all hell will break loose. Not only do we have to account for all the extra stomachs, we also have to take care of the environment, which means we're just a little stuck between a rock and a hard place. Add to that the tendency of people to either impose their ideas of what's right to eat on others (vegans, vegetarians, paleo-diet people and anti-GM groups), or the other bunch



SOURCE: ISTOCKPHOTO.COM

A genetically modified crop

of people who don't care how their diet impacts the planet (beef eaters we mean you...), and we have a recipe for disaster.

This is where technology has to step in to save the day. The most obvious technology is genetic modification (GM), where the DNA of plants and animals are changed to make them yield more food, cost less to maintain, and secure them from disease and pests.

Medical understanding of genomes, and of DNA has been increasing steadily, and with more genetic modification experiments or projects being green-lighted across the globe, food production is sure to get a shot in the arm.

Although many still question the ethics, an example of aquaculture (fish farms) using GM trials to try and prevent fish from falling sick, or making them resistant to fish parasites such as sea lice will go a long way in making seafood farms more efficient.

TASTY OPTIONS

<http://dgit.in/PDayFood>
<http://dgit.in/BistroV>
<http://dgit.in/GMOaqua>

Agriculture is not only benefitting from genetic modification, but also from improvements in weather prediction and more accurate rainfall reports.

Most importantly, we expect to see the first lab grown meat and dairy products to become available in 2017. While this may disgust even the meat eaters amongst us, lab grown meat has several advantages. It can be grown lean and healthy with less fat and it is much greener, more efficient and much more moral than raising an animal from a baby, feeding it all its life and then killing it for meat as a young adult.

13 SELF-DRIVING

When you think of self-driving cars, you automatically think Google. This is because, for a long time, Google has led the way in

self-driving research. However, recently, Uber announced a self-driving car test to be run in California. They were promptly told by the California Department of Motor Vehicles that they would need to apply for autonomous licenses for their cars, pay extra money and agree to rigorous self-driving regulations. Uber refused, because unlike with other self-driving cars, Uber has a driver present ready to take over in an instant. Of course, Uber lost the battle, and then promptly decided to shift to Arizona, where Governor Doug Ducey welcomed them with "open arms and wide open roads". Looks like there will be something

unique for a few Arizonan Uber riders to tell their friends about.

In October 2016 Tesla announced that from then on, all Tesla cars would be sold with full

hardware capability of being self-driving, including even the Model 3. Of course, the cheaper cars will have the hardware but won't have self-driving capability just yet. The more expensive cars will have all of the self-driving capabilities, and the data gathered from those cars will be analysed, and when

Tesla feel they are confident enough to, they can just send an over the air update and enable self-driving on the other vehicles. It doesn't have to be an all or nothing solution either, because features such as automatic emergency braking could be enabled first, then collision warning, and so on, depending on how much testing Tesla got done in real-world scenarios, and how confident it was of the system being perfect. Tesla claim that their self-driving cars are much safer than human drivers, despite 2016 being the year the first person died in a self-driving accident.

Tesla is still confident that it is far safer than human drivers, because it's only had one fatality in over 200 million km driven by self-driving Teslas, as opposed to US human drivers, who have a fatality every 150 million km.



SOURCE: FLICKR

Uber's self-driving car

As long as the second fatality happens much after 400 million miles have been self-driven by Teslas, this claim will hold. If the next fatality happens before 300 million km, Tesla might be in big trouble. Actually, the next fatality might be big trouble no matter how many million miles later it comes! This particular capability is under high levels of scrutiny and if you ask us, even the slightest mistake is a big deal.